Youngjoon Cheon

Research Associate, Korea University
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RESEARCH INTERESTS

Processor and Accelerator Architecture · Memory Systems · Hardware–Software Co-Design and Interfaces

EDUCATION

Korea University Sep. 2024 – Aug. 2026
M.S. in Electrical and Computer Engineering, GPA: 3.94/4.5 Seoul, South Korea
Thesis: *Segmentation-based Address Translation in GPUs for Accelerating LLM Inference*
Advisor: Prof. Jeongseob Ahn

Ajou University Mar. 2018 – Aug. 2024
B.S. in Software and Computer Engineering, GPA: 3.77/4.5 Suwon, South Korea
Minor in Mathematics

PUBLICATIONS

Youngjoon Cheon, Yunho Oh, and Jeongseob Ahn, "Towards Segmentation-based Address Translation for LLM Inference," *IEEE Computer Architecture Letters* (**CAL**), 2026. [**Paper**]

RESEARCH EXPERIENCE

Research Associate Oct. 2026 – Present
Computer Systems Laboratory, Korea University *Collaborator: Prof. Jeongseob Ahn*

Graduate Research Assistant Sep. 2024 – Aug. 2026
Computer Systems Laboratory, Korea University *Advisor: Prof. Jeongseob Ahn*

· **HW/SW Co-Design for LLM Address Translation** Jan. 2025 – May. 2026

- Characterized KV cache-induced TLB contention that evicted frequently reused model weight translations during long-context LLM inference
- Instrumented NVIDIA's UVM driver to verify physical contiguity of weight allocations in the evaluated setup
- Modified vLLM for virtually contiguous weight allocation and designed a GPU segmentation table to bypass TLB lookups for model weights
- Reduced L2 TLB miss rate from 91.5% to 17.0% and achieved 2.51× IPC over conventional paging in Accel-Sim

· **Intra-GPU Memory Tiering for KV Cache Scaling** Apr. 2024 – Dec. 2024

- Proposed an intra-GPU HBM–LPDDR hierarchy with layer-granular KV cache placement and migration
- Integrated GPGPU-Sim with Ramulator and modeled separate HBM and LPDDR memory controllers
- Implemented a simplified queue-based DMA copy engine model for data migration independent of SM execution

Undergraduate Research Intern Sep. 2022 – Aug. 2024
Computer Systems Laboratory, Ajou University *Advisor: Prof. Jeongseob Ahn*

· **Heterogeneous xPU–PIM Execution for LLM Inference** Jul. 2023 – Mar. 2024

- Modified the Hugging Face inference pipeline to offload attention and KV cache storage to an NVLink-connected GPU serving as a PIM surrogate
- Implemented inter-GPU tensor transfers and profiled synchronization and PyTorch API overheads using NVIDIA Nsight Systems
- Supported batch size 32 (single GPU out of memory); measured 3.4% higher end-to-end generation latency at batch size 16 (Llama-2 7B, 512 token input)

AWARDS AND HONORS

Academic Excellence Scholarship (30-50% Tuition Waivers) Fall 2019, Fall 2022, Fall 2023
Ajou University

TEACHING EXPERIENCE

Graduate Teaching Assistant, Korea University

· **Digital Systems and Design Laboratory** Fall 2024, Fall 2025

- Prepared and delivered weekly lectures on digital logic; designed and supervised Verilog/FPGA labs
- Developed exam questions and graded assignments and final projects

· **Computer Architecture** Spring 2025, Spring 2026

- Designed a programming assignment requiring students to implement a RISC-V assembler (**GitHub link**)

TECHNICAL SKILLS

Languages	Verilog, C/C++, CUDA, Python, RISC-V Assembly, Shell
Frameworks	PyTorch, vLLM
Simulators & Tools	GPGPU-Sim, Accel-Sim, Ramulator, DRAMSim2, Nsight Systems/Compute, Vivado, Git

PROFESSIONAL SERVICE

Artifact Evaluation Committee Member

· ACM Symposium on Operating Systems Principles (SOSP 2026)
· IEEE International Symposium on High-Performance Computer Architecture (HPCA 2026)